Notes

Introduction to Hybridization

Welcome and Revision

Hello everyone! We're continuing our journey in basic chemistry. In our previous session, we focused on understanding and drawing chemical structures. Today, we're going to delve into a crucial concept: **Hybridization**.

What is Hybridization?

Hybridization is a fundamental concept in chemistry that helps us understand the bonding and geometry of molecules. It involves the mixing of atomic orbitals to form new hybrid orbitals, which are more suitable for the formation of chemical bonds. This process allows atoms to achieve more stable configurations and explains the observed molecular shapes.

Key Concepts to Recall:

Before we dive into hybridization, it's important to have a clear understanding of:

- **Atomic Orbitals:** The regions around the nucleus where electrons are most likely to be found (s, p, d, f orbitals).
- **Valence Electrons:** The electrons in the outermost shell of an atom, which are involved in chemical bonding.
- **Lewis Structures:** A way to represent the valence electrons of atoms within a molecule, showing how they are bonded together.

If you have any problems understanding these basic concepts, please review them, as they form the foundation for understanding hybridization. We will be building upon our knowledge of drawing structures to explore how hybridization influences these structures.

What is Hybridization?

Hybridization is the intermixing of valence orbitals of the central atom. This process leads to the formation of new orbitals, which are known as hybrid orbitals. These hybrid orbitals possess the same shape and energies. The primary purpose of hybridization is to achieve a more stable and energetically favorable arrangement for bonding.

Introduction to Hybridization: BeCl2 Example

Understanding Hybridization

Hybridization involves the mixing of atomic orbitals of the same central atom to form new hybrid orbitals with equivalent energy and shape. We will explore different types of hybridization, starting

Notes

To understand hybridization, let's consider the example of Beryllium Dichloride (BeCl2).

- **Central Atom:** Beryllium (Be) is the central atom.
- **Electronic Configuration of Beryllium:** The ground state electronic configuration of Beryllium is $1s^2 2s^2$. The second shell is the valence shell, containing the $2s$ and $2p$ orbitals. The $2p$ orbital is empty in the ground state.
    - **Ground State:**
        - $1s^2$
        - $2s^2$ (two paired electrons)
        - $2p^0$ (empty)

Excitation of Beryllium

For bond formation, electrons need to be unpaired. In Be, both $2s$ electrons are paired. To unpair them, energy is supplied, causing one electron from the $2s$ orbital to get excited to the higher energy $2p$ orbital.

- **Excited State:**
    - $1s^2$
    - $2s^1$ (one electron)
    - $2p^1$ (one electron)

Now, Be has two singly filled orbitals ($2s$ and $2p$), which can participate in bonding.

SP Hybridization in BeCl2

Formation of SP Hybrid Orbitals

In the excited state, one s orbital and one p orbital of beryllium mix to form two new hybrid orbitals. This type of mixing is called **sp hybridization**.

- **Mixing:** One $2s$ orbital + One $2p$ orbital
- **Result:** Two new orbitals, called **sp hybrid orbitals**.

These sp hybrid orbitals have:

- The same shape.
- The same energy.
- Each sp hybrid orbital contains one electron.

- One larger lobe (typically considered positive).
- One smaller lobe (typically considered negative).

Orientation of SP Hybrid Orbitals

To minimize repulsion between the electrons in the hybrid orbitals, these two sp hybrid orbitals orient themselves as far apart as possible. This results in an orientation at **180 degrees** to each other, forming a linear arrangement.

- Beryllium has two sp hybrid orbitals, each with one electron, oriented at 180° to each other.

Bond Formation in BeCl2

Electronic Configuration of Chlorine

Now, let's consider the side atom, Chlorine (Cl).

- **Electronic Configuration of Chlorine:** The valence shell configuration of Chlorine is $3s^2 3p^5$. This means it has seven valence electrons.
  - $3s^2$ (two paired electrons)
  - $3p^5$ (two paired electrons in two orbitals, one singly occupied orbital)

Chlorine has one singly occupied p orbital (e.g., $3p_z^1$) that can participate in bond formation.

Overlap and Sigma Bond Formation

Each of the two sp hybrid orbitals of beryllium, each containing one electron, will overlap with a singly occupied p orbital from two different chlorine atoms.

1. **First Chlorine Atom:** One p orbital from the first chlorine atom overlaps axially with one sp hybrid orbital of beryllium. This axial overlap forms a **sigma (σ) bond**.
2. **Second Chlorine Atom:** Similarly, one p orbital from the second chlorine atom overlaps axially with the other sp hybrid orbital of beryllium, forming another **sigma (σ) bond**.

Thus, two sigma bonds are formed in BeCl2.

Geometry and Bond Angle of BeCl2

Molecular Geometry

With two sigma bonds formed, one in each direction from the central beryllium atom, the molecule adopts a specific geometry.

Notes

Summary of SP Hybridization Characteristics

For sp hybridization:

- **Bond Angle:** 180°
- **Geometry:** Linear

Electronic Configuration of Boron

Boron's Electronic Configuration

Boron is a p-block element. Its outer electronic configuration is $2s^2 2p^1$. This means it has two electrons in the $2s$ orbital and one electron in the $2p$ orbital.

- **Ground State:**
  - $2s$: [↑↓]
  - $2p$: [↑][ ][ ]

Excitation of Boron

To form three bonds, boron requires three singly filled orbitals. In its ground state, it only has one singly filled orbital. Therefore, an electron from the $2s$ orbital is excited to a higher energy $2p$ orbital.

- **Excited State:**
  - $2s$: [↑]
  - $2p$: [↑][↑][ ]

This excitation provides three singly filled orbitals, which are now ready for hybridization.

sp2 Hybridization

Hybridization Process

After excitation, the one $2s$ orbital and two $2p$ orbitals intermix to form new hybrid orbitals. This process is called hybridization.

- **Type of Hybridization:** Since one s orbital and two p orbitals are involved, the hybridization is sp^2.

Characteristics of sp2 Hybrid Orbitals

- **Number of Hybrid Orbitals:** Three new sp^2 hybrid orbitals are formed.
- **Shape and Energy:** These three hybrid orbitals are identical in shape and energy.

- **Shape of Hybrid Orbitals:** Each hybrid orbital has a large lobe and a small lobe, with one electron in the larger lobe.

Geometry of sp2 Hybridization

When three hybrid orbitals are oriented at 120° in a plane, they form a trigonal planar geometry.

Electronic Configuration of Fluorine

Fluorine's Electronic Configuration

Fluorine's outer electronic configuration is $2s^2 2p^5$. This means it has two electrons in the $2s$ orbital and five electrons in the $2p$ orbital.

- **Ground State:**
  - $2s$: [↑↓]
  - $2p$: [↑↓][↑↓][↑]

Fluorine has one singly filled $2p$ orbital, which allows it to form one bond.

Formation of Boron Trifluoride (BF3)

Overlap of Orbitals

In BF_3, boron forms three bonds with three fluorine atoms. Each of boron's three sp^2 hybrid orbitals (each containing one electron) overlaps with the singly filled $2p$ orbital of a fluorine atom (also containing one electron).

Sigma Bonds

- **Head-to-Head Overlap:** The overlap between the sp^2 hybrid orbital of boron and the $2p$ orbital of fluorine occurs head-to-head along the internuclear axis.
- **Resulting Bonds:** This type of overlap forms sigma (σ) bonds.
- **Number of Sigma Bonds:** Three sigma bonds are formed in BF_3.

Molecular Geometry and Bond Angle

- **Bond Angle:** Due to the sp^2 hybridization of boron, the three $B-F$ sigma bonds are oriented at 120° to each other.
- **Geometry:** The molecule adopts a trigonal planar geometry, with boron at the center and the three fluorine atoms at the corners of an equilateral triangle.

Summary of sp2 Hybridization in BF3

- **Hybridization:** sp^2

Notes

Introduction to sp3 Hybridization in CH4

Central Atom and Electronic Configuration

In methane (CH4), carbon is the central atom as hydrogen cannot be a central atom. The electronic configuration of carbon is $1s^2 2s^2 2p^2$. This means:

- The $2s$ orbital has two paired electrons.
- The $2p$ orbital has two unpaired electrons (one in each of two p sub-orbitals).

Excitation of Electrons

For bonding, carbon needs four unpaired electrons. Therefore, one electron from the $2s$ orbital gets excited and jumps to an empty $2p$ orbital. After excitation, the configuration becomes:

- $2s$ orbital: one electron
- $2p$ orbital: three electrons (one in each of the three p sub-orbitals)

Formation of sp3 Hybrid Orbitals

All four orbitals (one $2s$ and three $2p$) participate in hybridization. This process is called sp^3 hybridization. After hybridization, four new hybrid orbitals are formed. These sp^3 hybrid orbitals are identical in shape and energy. Each sp^3 hybrid orbital contains one electron, ready to form a bond.

Geometry and Orientation of sp3 Hybrid Orbitals

Non-Planar Geometry

With four hybrid orbitals, the geometry will be non-planar. The electron pairs around the central atom repel each other, and to minimize these repulsions, the electron pairs (and thus the hybrid orbitals) arrange themselves as far apart as possible.

Tetrahedral Geometry

This arrangement leads to a tetrahedral geometry. The bond angle between any two sp^3 hybrid orbitals is $109^\circ 28'$. This specific angle maximizes the distance between the electron pairs, thereby minimizing repulsion.

Visualizing the Tetrahedral Structure

Imagine the carbon atom at the center. The four sp^3 hybrid orbitals are oriented as follows:

- One orbital points vertically upwards.

The fourth orbital points out of the plane, forming an angle of 109.28° with the other three. This is a 3D structure.

Bonding with Hydrogen

Each of these four sp^3 hybrid orbitals will overlap with a $1s$ orbital from a hydrogen atom (hydrogen's configuration is $1s^1$) to form four sigma (σ) bonds. Each bond involves the sharing of two electrons, forming a covalent bond.

Representing 3D Geometry: Wedges

Drawing the Tetrahedral Structure

To represent the 3D tetrahedral geometry on a 2D surface, specific conventions are used:

- **Vertical Bond:** One C-H bond is often drawn vertically.
- **In-Plane Bonds:** Two C-H bonds can be drawn in the plane of the paper.
- **Solid Wedge:** A solid wedge (thickened line) represents a bond pointing **towards the observer** (out of the plane of the paper). For example, if carbon is in the plane, one hydrogen bond comes out towards you.
- **Dashed Wedge:** A dashed wedge (broken line) represents a bond pointing **away from the observer** (into the plane of the paper). For example, the fourth hydrogen bond goes away from you, into the plane.

Resultant Geometry

When the corners of this arrangement are joined, a tetrahedron is formed. Carbon is at the center, and the hydrogen atoms are at the corners. Therefore, sp^3 hybridization always results in a tetrahedral geometry.

Hybridization and its Characteristics

Introduction to Hybridization

Hybridization is the concept of mixing atomic orbitals to form new hybrid orbitals suitable for the pairing of electrons to form chemical bonds. We will discuss different types of hybridization, their geometries, bond angles, and an important concept called percentage S character.

SP Hybridization

- **Formation:** One S orbital mixes with one P orbital.
- **Geometry:** Linear
- **Bond Angle:** 180°

SP2 Hybridization

- **Formation:** One S orbital mixes with two P orbitals.
- **Geometry:** Trigonal Planar
- **Bond Angle:** 120°
- **Percentage S Character:** 33.33%
  - For SP2, it's $(1/3) \times 100 = 33.33\%$. This means there is 33.33% S character and 66.67% P character.

SP3 Hybridization

- **Formation:** One S orbital mixes with three P orbitals.
- **Geometry:** Tetrahedral
- **Bond Angle:** $109^{\circ}28^{'}$
- **Percentage S Character:** 25%
  - For SP3, it's $(1/4) \times 100 = 25\%$. This means there is 25% S character and 75% P character.

Calculation of Percentage S Character

The percentage S character is calculated using the formula: $\text{Percentage S character} = \frac{\text{Number of S orbitals}}{\text{Total number of orbitals}} \times 100$

Relationship between S Character and Bond Angle

There is a direct relationship between the percentage S character and the bond angle:

- As the percentage S character increases, the bond angle also increases.
  - SP3 (25% S character) has a bond angle of $109^{\circ}28^{'}$.
  - SP2 (33.33% S character) has a bond angle of 120°.
  - SP (50% S character) has a bond angle of 180°.
- **Key Principle:** Greater the amount of S character in the hybrid orbitals, greater will be the bond angle.
- Conversely, as the percentage P character decreases (which happens when S character increases), the bond angle increases. This implies an inverse relationship between bond angle and P character.

Hybridization Involving d-Orbitals: PCl5 Example

phosphorus pentachloride (PCl_5).

Electronic Configuration of Phosphorus (P)

- **Central Atom:** Phosphorus (P)
- **Atomic Number:** 15
- **Electronic Configuration:** $1s^2 2s^2 2p^6 3s^2 3p^3 3d^0$
  - The third shell ($n = 3$) contains 3s, 3p, and 3d orbitals. In the ground state, the 3d orbitals are empty.
  - The valence shell (outermost shell) is the third shell, containing 5 electrons ($3s^2 3p^3$).

Excitation of Electrons

For bond formation, electrons from the valence shell participate. In PCl_5, phosphorus needs to form five bonds with five chlorine atoms. To achieve this, one electron from the 3s orbital gets excited to an empty 3d orbital within the same shell.

- **Ground State Configuration (Valence Shell):**
  - 3s: ↑↓
  - 3p: ↑ _ ↑ _ ↑ _
  - 3d: _____
- **Excited State Configuration (Valence Shell):**
  - One electron from 3s moves to 3d.
  - 3s: ↑ _
  - 3p: ↑ _ ↑ _ ↑ _
  - 3d: ↑ ____

sp^3d Hybridization

In the excited state, phosphorus has five half-filled orbitals (one 3s, three 3p, and one 3d). These five atomic orbitals hybridize to form five new equivalent hybrid orbitals. This type of hybridization is called sp^3d hybridization.

- **Participating Orbitals:** One s-orbital, three p-orbitals, and one d-orbital.
- **Result:** Five sp^3d hybrid orbitals.

Electronegativity and s-Character

- **sp^3 Hybrid Carbon:** 25% s-character
- **sp^2 Hybrid Carbon:** 33.33% s-character
- **sp Hybrid Carbon:** 50% s-character

Order of Electronegativity: $sp > sp^2 > sp^3$ Thus, sp hybrid carbon is the most electronegative among these three.

Bond Strength and s-Character

Bond strength is also directly proportional to the s-character. Greater the s-character, stronger will be the bonds.

Characteristics of s and p Orbitals

- **s-orbital:** Smaller, spherical, bulky (thick).
- **p-orbital:** Longer in length, thinner.

Effect of s-Character on Hybrid Orbital Shape

When s-character increases in a hybrid orbital, the hybrid orbital starts to resemble an s-orbital more. This means:

- It will decrease in length.
- It will become more bulky/spherical.

Conversely, if p-character increases, the hybrid orbital becomes longer and thinner, resembling a p-orbital more.

Bond Length and Bond Strength Relationship

- **Bond Length:** The distance between the centers of two bonded atoms.
- **Bond Strength:** The energy required to break a bond.

Consider three hybrid orbitals with varying s-character:

1. **High s-character:** More spherical, shorter lobe.
2. **Medium s-character:** Intermediate shape and length.
3. **Low s-character (high p-character):** Longer, thinner lobe.

When these orbitals form bonds:

- The bond formed by the orbital with **high s-character** will be **shorter** because the electron pair is closer to the nucleus, leading to stronger attraction.

Conclusion:

- Greater the bond length, weaker is the bond.
- Shorter the bond length, stronger is the bond.

Therefore, as s-character increases:

- Bond length decreases.
- Bond strength increases.

Summary of Properties Dependent on s-Character:

- Electronegativity $\propto$ s-character
- Bond strength $\propto$ s-character
- Bond length $\propto$ 1/s-character

Geometry of sp^3d Hybridization: Trigonal Bipyramidal

Geometry of sp^3d Hybridization

For sp^3d hybridization, the geometry is **trigonal bipyramidal (TBP)**.

In PCl_5, phosphorus forms five bonds using its five sp^3d hybrid orbitals.

Structure of Trigonal Bipyramidal Geometry

- **Equatorial Bonds:** Three bonds lie in a plane, forming a triangle around the central atom. These are called equatorial bonds.
  - The bond angle between any two equatorial bonds is 120°.
- **Axial Bonds:** Two bonds are perpendicular to the equatorial plane, one above and one below. These are called axial bonds.
  - The bond angle between an axial bond and any equatorial bond is 90°.
  - The bond angle between the two axial bonds is 180°.

Types of Bonds in sp^3d Hybridization

1. **Equatorial Bonds:** Three bonds in the planar triangle.
2. **Axial Bonds:** Two bonds along the axis perpendicular to the equatorial plane.

Bond Angles in Trigonal Bipyramidal Geometry

- **Equatorial-Equatorial:** 120°
- **Equatorial-Axial:** 90°

Notes

Importance of sp3d2 Hybridization

This hybridization is crucial as it is frequently encountered in coordination compounds.

Example: Sulfur Hexafluoride (SF6)

- **Central Atom:** Sulfur (S)
- **Atomic Number of Sulfur:** 16
- **Electronic Configuration:** 2, 8, 6
- **Valence Shell Configuration:** Sulfur has six valence electrons in its third shell.
  - The third shell begins with the 3s orbital.
  - The configuration is $3s^2 3p^4 3d^0$.

Orbital Representation of Sulfur (Ground State)

- **3s orbital:** Contains two electrons.
- **3p orbitals:** Contains four electrons (one paired, two unpaired).
  - $3p_x^2 3p_y^1 3p_z^1$
- **3d orbitals:** Empty (contains zero electrons).

Excitation of Sulfur Atom

To form six bonds in SF6, sulfur needs six unpaired electrons. This requires excitation of electrons from the 3s and 3p orbitals to the empty 3d orbitals.

- One electron from the 3s orbital is excited to a 3d orbital.
- One electron from a 3p orbital is excited to another 3d orbital.

Orbital Representation of Sulfur (Excited State)

After excitation (denoted by a star on the sulfur atom):

- **3s orbital:** Contains one electron.
- **3p orbitals:** Contains three electrons (all unpaired).
- **3d orbitals:** Contains two electrons (both unpaired).

Now, there are six unpaired electrons available for bonding.

sp3d2 Hybridization Process and Characteristics

Intermixing of Orbitals

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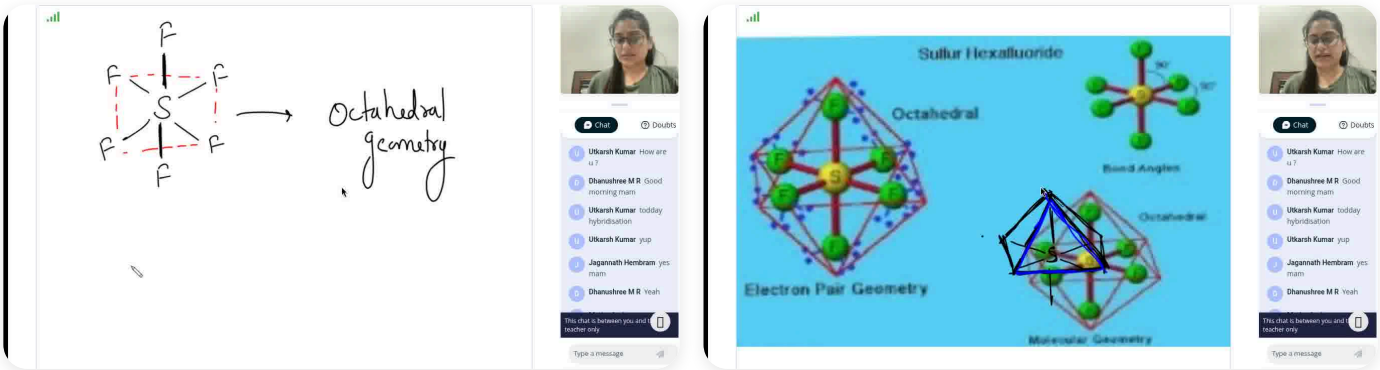
Characteristics of sp3d2 Hybrid Orbitals

- **Number of Orbitals:** Six new hybrid orbitals are formed.
- **Shape and Geometry:** All six new orbitals have identical shape and energy.
- **Electron Occupancy:** Each of these six hybrid orbitals contains one electron, ready to form a bond.
- **Conservation of Electrons:** The total number of electrons (paired or unpaired) remains constant before and after hybridization.

Orientation of sp3d2 Hybrid Orbitals

- **Bond Angles:** All hybrid orbitals are oriented at 90° to each other, with opposite bonds at 180°.
- **Absence of Axial/Equatorial Positions:** Unlike trigonal bipyramidal geometry, there are no distinct axial and equatorial positions in sp3d2 hybridization.
- **Geometry:** The resulting geometry is **octahedral**.

Octahedral Geometry in SF6



Formation of Bonds in SF6

Sulfur, as the central atom, forms six bonds with six fluorine atoms using its six sp3d2 hybrid orbitals.

Visualizing Octahedral Geometry

Imagine the sulfur atom at the center of a coordinate system:

- **Axes:** The six bonds are directed along the positive and negative x, y, and z axes.
  - Two bonds along the z-axis (one positive, one negative).
  - Two bonds along the x-axis (one positive, one negative).
  - Two bonds along the y-axis (one positive, one negative).

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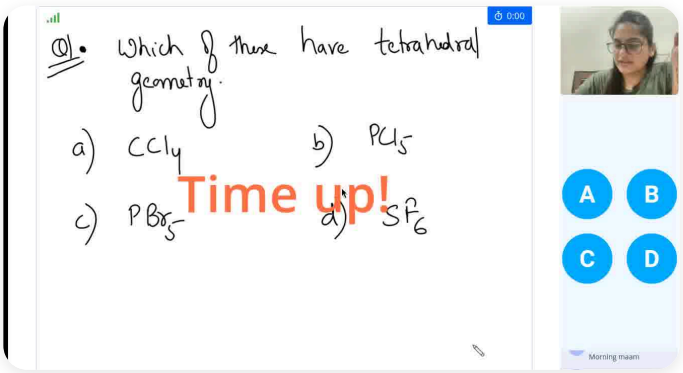
An octahedron is a polyhedron with eight faces. In this molecular geometry:

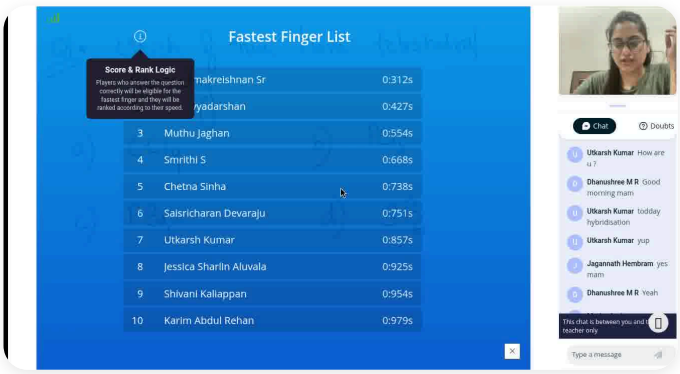
- **Triangular Faces:** There are eight triangular faces formed by connecting the fluorine atoms.
  - Four triangular faces are in the upward direction.
  - Four triangular faces are in the downward direction.

Key Takeaway for Octahedral Geometry

In octahedral geometry, bonds are formed along the X, Y, and Z axes.

Quiz 1: Tetrahedral Geometry





Question

Which of these have tetrahedral geometry? a) CCl_4 b) PCl_5 c) PBr_5 d) SF_6

Solution

- CCl_4 is sp^3 hybridized and has a tetrahedral geometry.
- PCl_5 is sp^3d hybridized.
- PBr_5 is sp^3d hybridized.
- SF_6 is octahedral.

Therefore, the correct answer is a) CCl_4.

Quiz 2: Non-Linear Molecules

Question

Which of these is not linear?

Solution

- BeF_2 is linear.

- BF_3 is sp^2 hybridized and has a trigonal planar geometry, which is not linear.

Therefore, the correct answer is BF_3.

Difference Between Sigma (σ) and Pi (π) Bonds

Formation

- **Sigma (σ) Bond:** Formed by head-on (axial) overlap of atomic orbitals.
- **Pi (π) Bond:** Formed by lateral (sideways) overlap of atomic orbitals.

Strength

- **Sigma (σ) Bond:** Stronger than pi bonds due to greater overlap.
- **Pi (π) Bond:** Weaker than sigma bonds.

Existence

- **Sigma (σ) Bond:** Can exist alone (independently).
- **Pi (π) Bond:** Always exists with a sigma bond; cannot exist alone.

Orbital Involvement

- **Sigma (σ) Bond:** Mostly formed by hybrid orbitals.
- **Pi (π) Bond:** Always formed by unhybridized p or d orbitals.

Role in Geometry

- **Sigma (σ) Bond:** Decides the geometry of the molecule. The number of sigma bonds helps determine the molecular geometry.
- **Pi (π) Bond:** Has no role in deciding the geometry of the molecule; they are present along with sigma bonds.

Number of Bonds Between Two Atoms

- **Sigma (σ) Bond:** There is always only one sigma bond between any two atoms. A sigma bond is formed first and is unique.
- **Pi (π) Bond:** Can be multiple pi bonds between two atoms (e.g., in double or triple bonds).

Summary of Bond Types

- **Single Bond:** Always one sigma (σ) bond.
- **Double Bond:** One sigma (σ) bond and one pi (π) bond.

Notes

Hydrogen atoms always form sigma bonds.

Counting Sigma and Pi Bonds - Example 1 ($CH_2 = CH_2$)

Molecule

$CH_2 = CH_2$

Structure

```
 H    H
|    |
C = C
|    |
H    H
```

Counting Sigma Bonds

1. C-H bonds: 4 sigma bonds (one for each H)
2. C-C single bond equivalent in double bond: 1 sigma bond

Total Sigma Bonds = $4 + 1 = 5$

Counting Pi Bonds

1. C=C double bond contains one pi bond.

Total Pi Bonds = 1

Counting Sigma and Pi Bonds - Example 2 (Complex Hydrocarbon)

Molecule

Consider a molecule with the following structure:

```
 H    H    H
|    |    |
C - C = C - C ≡ C - H
|    |    |
H    H    H
```

Counting Sigma Bonds

1. C-H bonds: 8 sigma bonds (3 on first C, 2 on second C, 1 on third C, 1 on fourth C, 1 on fifth C)

4. C≡C triple bond: 1 sigma bond

Total Sigma Bonds = 8 + 2 + 1 + 1 = 12

Let's re-evaluate based on the lecture's counting method:

- Single bonds are always sigma.
- Double bonds have one sigma and one pi.
- Triple bonds have one sigma and two pi.

Let's count the bonds in the structure provided in the lecture (which was slightly different from the one I inferred, but the principle is the same):

Assume the structure is:

```
 H   H   H   H
|   |   |   |
C - C = C - C ≡ C - H
|   |   |   |
H   H   H   H
```

Let's count the sigma bonds as per the lecture's example:

1. C-H bonds: 3 (leftmost C) + 1 (next C) + 1 (next C) + 1 (next C) + 1 (rightmost C) = 7 sigma bonds.
2. C-C single bond: 1 sigma bond.
3. C=C double bond: 1 sigma bond.
4. C≡C triple bond: 1 sigma bond.

Total sigma bonds = 7 + 1 + 1 + 1 = 10

However, the lecture states 14 sigma bonds. This implies a different structure or a more complex counting. Let's re-examine the lecture's counting for 14 sigma bonds:

If we consider a structure like:

```
       H   H   H
      |   |   |
H - C - C = C - C ≡ C - H
      |   |   |   |
      H   H   H   H
```

This would be a 5-carbon chain. Let's count:

- C-H bonds: 3 (leftmost C) + 1 (next C) + 1 (next C) + 1 (next C) + 1 (rightmost C) = 7 sigma bonds.
- C-C single bonds: 1 (between C1-C2) + 1 (between C3-C4) = 2 sigma bonds.

Total sigma bonds = $7 + 2 + 1 + 1 = 11$ sigma bonds. Still not 14.

The lecture's example for 14 sigma bonds must have been a larger or different molecule. Assuming the lecture's count of 14 sigma bonds and 3 pi bonds for a given structure:

Counting Pi Bonds

1. C=C double bond: 1 pi bond.
2. C≡C triple bond: 2 pi bonds.

Total Pi Bonds = $1 + 2 = 3$

This part of the counting is consistent with the rules for double and triple bonds.

Introduction to Ethylene (Ethene) and Hybridization

Understanding the Structure of Ethene ($CH_2 = CH_2$)

Let's consider the compound ethene, $CH_2 = CH_2$. In this molecule, we have two central carbon atoms, which we can label as C1 and C2.

Electron Configuration of Carbon

Any carbon atom has the electron configuration $1s^2 2s^2 2p^2$. For the valence shell, this means:

- Two electrons in the $2s$ orbital.
- Two electrons in the $2p$ orbitals.

Excitation of Carbon Atoms

Before bonding, carbon undergoes excitation. An electron from the $2s$ orbital is promoted to an empty $2p$ orbital. This results in:

- One electron in the $2s$ orbital.
- Three electrons in the $2p$ orbitals.

Determining Hybridization in Ethene

To determine the hybridization of each carbon atom in ethene, we need to count the number of sigma bonds formed by each carbon. Each carbon atom in ethene forms three sigma bonds:

- Two sigma bonds with hydrogen atoms.
- One sigma bond with the other carbon atom.

sp^2 Hybridization

In sp^2 hybridization:

- One s orbital and two p orbitals mix to form three sp^2 hybrid orbitals.
- These three sp^2 hybrid orbitals are oriented in a trigonal planar geometry, with bond angles of approximately 120°.
- One p orbital remains unhybridized and is perpendicular to the plane of the sp^2 hybrid orbitals.

Both C1 and C2 undergo sp^2 hybridization. Each carbon atom will have three sp^2 hybrid orbitals and one unhybridized p orbital.

Formation of Sigma Bonds in Ethene

Overlap of sp^2 Hybrid Orbitals

Each sp^2 hybrid orbital contains one electron. The formation of sigma bonds occurs through the head-on (axial) overlap of these hybrid orbitals.

1. **Carbon-Carbon Sigma Bond:** One sp^2 hybrid orbital from C1 overlaps head-on with one sp^2 hybrid orbital from C2. This forms a strong carbon-carbon sigma (σ) bond.
2. **Carbon-Hydrogen Sigma Bonds:** Each carbon atom has two remaining sp^2 hybrid orbitals. These orbitals overlap head-on with the $1s$ orbital of two hydrogen atoms. Hydrogen has an electron configuration of $1s^1$.
    - C1 forms two C-H sigma bonds.
    - C2 forms two C-H sigma bonds.

In total, there are five sigma bonds in ethene:

- One C-C sigma bond.
- Four C-H sigma bonds.

All these sigma bonds are formed by the head-on overlap of hybrid orbitals (or a hybrid orbital and an s orbital).

Formation of Pi Bonds in Ethene

Role of Unhybridized p Orbitals

After the formation of sigma bonds, each carbon atom still possesses one unhybridized p orbital. These unhybridized p orbitals:

- Each contains one electron.

Sideways Overlap and Pi Bond Formation

The unhybridized p orbital of C1 overlaps sideways with the unhybridized p orbital of C2. This sideways overlap leads to the formation of a pi (π) bond.

- The pi bond electron density is located above and below the plane of the sigma bond framework.
- A double bond consists of one sigma bond and one pi bond.

Summary of Bonding in Ethene

- **Double Bond:** The carbon-carbon double bond in ethene is composed of one sigma bond (from $sp^2 - sp^2$ overlap) and one pi bond (from $p - p$ sideways overlap).
- **Geometry:** The sp^2 hybridization results in a trigonal planar geometry around each carbon atom, with bond angles of approximately 120°. The entire molecule is planar.

It is important to remember that between any two atoms, there can only be one sigma bond. Any additional bonds between those two atoms must be pi bonds.

Sigma and Pi Bonds in Allyl Cyanide

Understanding Sigma and Pi Bonds

- A sigma (σ) bond is represented by a single line in a chemical structure.
- A pi (π) bond is also represented by a line, but it is formed *after* a sigma bond between the same two atoms.

Counting Sigma and Pi Bonds in Allyl Cyanide

Let's analyze the structure of allyl cyanide to count the sigma and pi bonds. Allyl cyanide has the following structure:

$$CH_2 = CH - C \equiv N$$

Breaking down the bonds:

- **C=C (double bond):** Contains one sigma (σ) bond and one pi (π) bond.
- **C-C (single bond):** Contains one sigma (σ) bond.
- **C≡N (triple bond):** Contains one sigma (σ) bond and two pi (π) bonds.
- **C-H (single bonds):** Each contains one sigma (σ) bond.

Let's count them systematically:

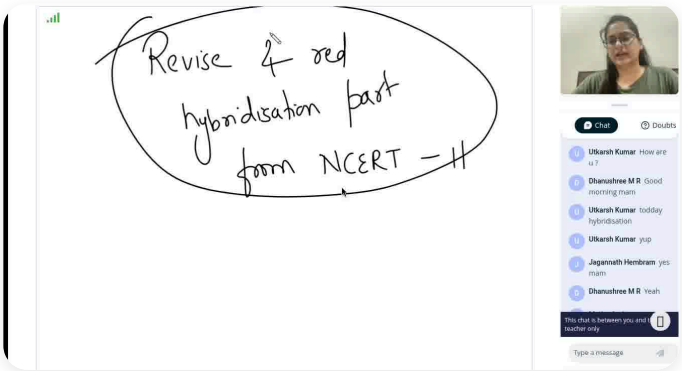
2. **CH group:**
   - One C-H single bond: 1 sigma bond.
   - One C-C single bond (connecting to the C≡N group): 1 sigma bond.
3. **C≡N group:**
   - One C≡N triple bond: 1 sigma bond, 2 pi bonds.

Total Count:

- **Sigma (σ) bonds:** 2 (from CH_2) + 1 (from C=C) + 1 (from CH) + 1 (from C-C) + 1 (from C≡N) = **6 sigma bonds**.
- **Pi (π) bonds:** 1 (from C=C) + 2 (from C≡N) = **3 pi bonds**.

This type of question, involving counting sigma and pi bonds in a given structure, has appeared in previous JEE Advanced examinations.

Homework and Next Steps



Homework Assignment

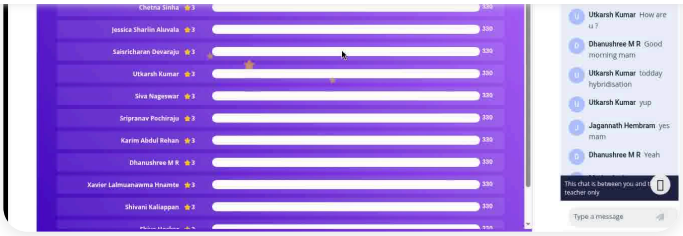
- **Revise:** Go through today's class content on hybridization.
- **Read:** Study the hybridization section from the NCERT Class 11 textbook.

Next Class Topic

- The next topic to be covered will be **Coordination Compounds**. This topic will become easier to understand after gaining a strong foundation in organic chemistry.

Leaderboard and Conclusion

Notes





Leaderboard

Congratulations to the top performers in today's session:

- Jaya Jayaram
- B. Divya Darshan
- Muthu
- Chetna
- Jessica
- Sain
- Sree Charan
- Utkarsh
- Siva Sain
- Pranav
- Karim
- Dhanushree
- Xavier
- Shivani
- Shiva
- Jagannath

Session Summary

Today's session focused on understanding sigma and pi bonds and their application in determining the bonding in molecules like allyl cyanide. The concept of hybridization was also reinforced. Students are encouraged to complete their homework and prepare for the next class.